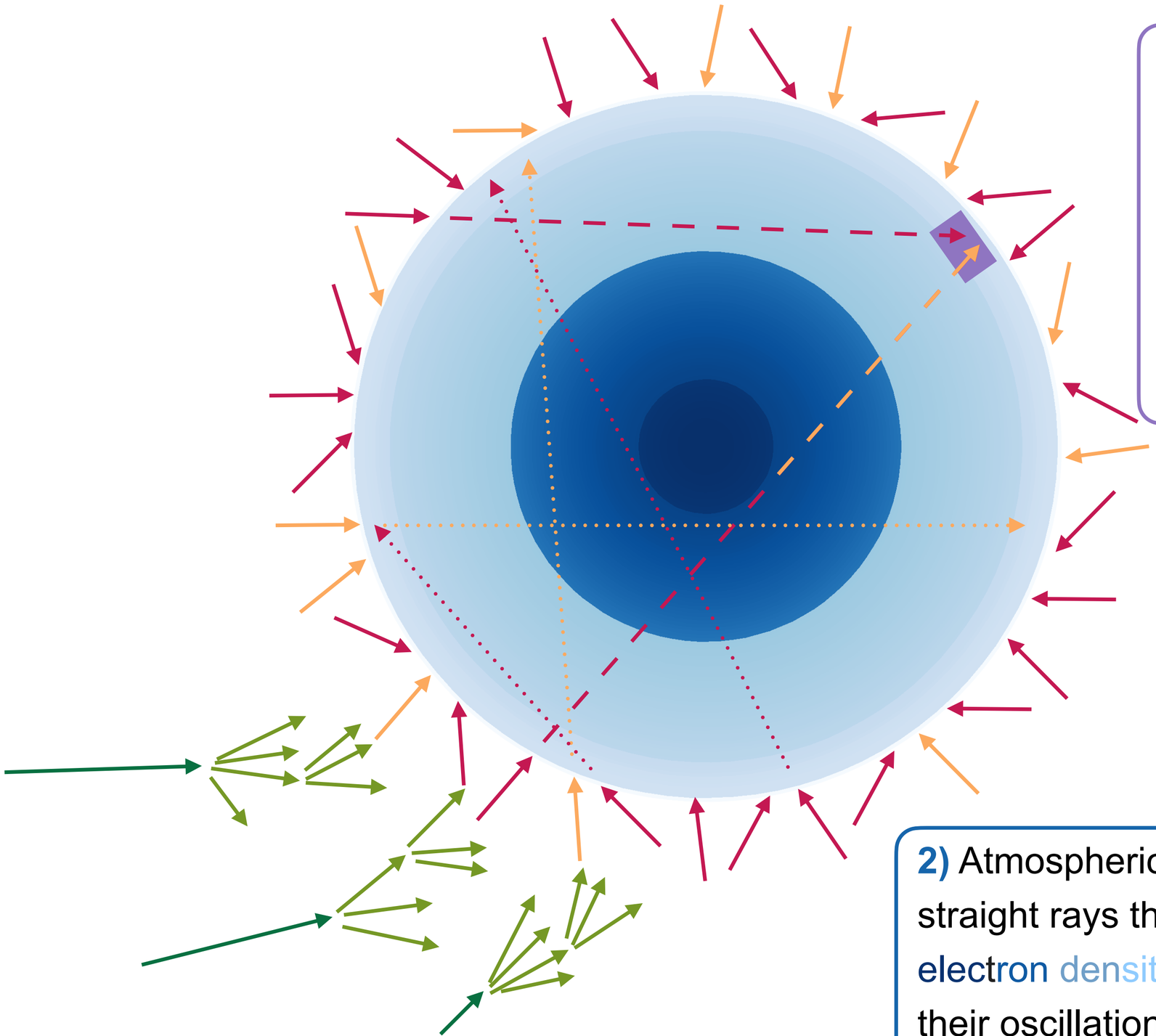


3) Finally, part of the oscillated atmospheric neutrino flux reaches a neutrino detector. The third component of the forward model accounts for the detector's efficiency and measurement resolution.

2) Atmospheric neutrinos travel along straight rays through the Earth. The electron density along their paths affects their oscillation behaviour. The second component of the forward model describes neutrino oscillations for a given Earth model.

1) Cosmic rays, produced in violent events across the universe, enter our atmosphere and generate showers of secondary particles, including neutrinos. The first component of the forward model is a description of this atmospheric neutrino flux.